

Comparative Analysis of Machine Learning Algorithms for Earthquake Magnitude Classification in Muğla - **Data Preparation**

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```
# reticulate::py_require("pandas") for R-to-Python integration ,  
  
# please run in the R console  
# reticulate::source_python("earthquake.py")
```

```
import pandas as pd  
  
# import mugla.txt data  
mugla = pd.read_csv(  
    "mugla.txt",  
    sep=r"\s+",  
    header=0,  
    index_col=0  
)  
  
mugla.head()  
  
print(mugla.head())
```

	id	date	time	lat	long	country	city	\
6	2.010000e+13	2005.01.21	12:04:03 AM	37.11	27.75	turkey	mugla	
12	1.990000e+13	1989.04.27	12:45:19 AM	37.04	28.04	turkey	mugla	
30	1.970000e+13	1967.05.22	12:46:02 AM	36.59	29.35	turkey	mugla	
49	2.000000e+13	1996.02.18	12:03:13 AM	37.17	27.64	turkey	mugla	
80	1.990000e+13	1989.01.02	12:25:46 AM	37.05	27.40	turkey	mugla	

		area	dist	depth	...	mw	ms	mb	mg	day	month	year	\
6	demirciler_milas	0.1	32.8	...	NaN	0.0	0.0	0	0	21	1	2005	
12	kultak_milas	0.1	9.0	...	NaN	0.0	0.0	0	0	27	4	1989	
30	cobanlar_fethiye	0.2	54.0	...	4.9	4.6	4.6	0	0	22	5	1967	
49	dorttepe_milas	0.2	5.0	...	NaN	0.0	0.0	0	0	18	2	1996	
80	konacik_bodrum	0.2	9.0	...	NaN	0.0	0.0	0	0	2	1	1989	

	day_diff	dirname	dir
6	30946	west	4
12	25198	south	3
30	17187	west	4
49	27686	south	3
80	25083	east	2

[5 rows x 23 columns]

```
# total NA values from all variables
print("\nMissing values:")
print(mugla.isna().sum())
```

Missing values:

id	0
date	0
time	0
lat	0
long	0
country	0
city	0
area	1
dist	0
depth	0
xm	0
md	0
richter	0
mw	554
ms	0
mb	0
mg	0
day	0
month	0
year	0

```
day_diff      0
dirname       0
dir           0
dtype: int64
```

```
# splitting of data into train and test data

from sklearn.model_selection import train_test_split

train_data, test_data = train_test_split(
    mugla,
    test_size=0.20,
    random_state=123,
    stratify=mugla["mg"]      # stratified split
)

print("Train dataset:", train_data.shape)
```

Train dataset: (532, 23)

```
X_train = train_data.drop(columns=["mg"])
y_train = train_data["mg"]

X_test = test_data.drop(columns=["mg"])
y_test = test_data["mg"]
print("Test dataset:", test_data.shape)
```

Test dataset: (134, 23)

```
# check class proportions
print("Train class proportions:")
print(y_train.value_counts(normalize=True))

print("\nTest class proportions:")
print(y_test.value_counts(normalize=True))
```

Train class proportions:

```
mg
0    0.947368
1    0.052632
```

```
Name: proportion, dtype: float64
```

```
Test class proportions:
```

```
mg
```

```
0    0.947761
```

```
1    0.052239
```

```
Name: proportion, dtype: float64
```